

Education

H.B.Sc., University of Toronto (Statistics, Astronomy & Astrophysics, Mathematics) 2018-2022

Research Experience

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- 2021 May - 2021 Aug **Undergraduate Summer Research Fellowship**
Canadian Institute for Theoretical Astrophysics
Continued previous work on the mass distribution of the Milky Way. Collaborated with members of the [H3 Survey](#) and published a first-author paper in *ApJ*.
Began a project with [Josh Speagle](#) using normalizing flows to make probabilistic predictions of the ages of stars.
Began a project with Norm Murray using Stan to fit dynamical models to geological data from thermal tides.
- 2020 Aug - 2021 May **Undergraduate Research Course**
Canadian Institute for Theoretical Astrophysics
Supervised by [Prof. Gwendolyn Eadie](#) and [Prof. Norman Murray](#). Study of mass distribution of the Milky Way in a Bayesian framework. Implemented a hierarchical Bayesian model for modelling the mass of galactic halos using a gradient-based sampling algorithm, increasing computational efficiency by two orders of magnitude over existing code.
- 2020 May - 2020 Aug **Summer Undergraduate Research Program**
Dunlap Institute for Astronomy and Astrophysics, University of Toronto
Supervised by [Prof. Allison Man](#). Study of molecular gas in a strongly-lensed galaxy group at $z = 2.9$. Analyzed ALMA data to infer molecular gas masses in several high-redshift galaxies. Resulted in a first-author publication.

Refereed Publications

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2. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han. “The Mass of the Milky Way from the H3 Survey”, submitted to *ApJ*.
 1. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. “Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$ ”, accepted for publication in *ApJ*. <https://arxiv.org/abs/2105.11572>

Honours and Awards

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- 2021 May - 2021 Aug **Undergraduate Summer Research Fellowship, (\$8396 CAD)**
Canadian Institute for Theoretical Astrophysics, University of Toronto
- 2020 May - 2020 Aug **Summer Undergraduate Research Program Award (\$9,500 CAD)**
Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Presentations

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- 2021 August **SDSS 2021 Collaboration Meeting, JHU/Online**
Lightning talk: “Predicting the ages of stars with machine learning.”
- 2021 May **Stellar Stats Workshop, UofT/Online**
Lightning talk: “Estimating the mass distribution of the Milky Way with Bayesian multi-level models.”
- 2021 May **Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online**
Poster presentation and lightning talk: “Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$.”

- 2021 Apr **Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online**
Presentation: “Molecular gas in gravitationally lensed galaxies at $z = 2.9$.”
- 2020 Oct **REQUIEM Galaxies Telecon, Online**
Presentation: “Molecular gas in gravitationally lensed galaxies at $z = 2.9$.”
- 2020 Aug **Summer Undergraduate Research Program Poster Session, UofT/Online**
Poster presentation: “Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$.”
Awarded prize for one of top three posters.

Media

- 2021 Jul **National Radio Astronomy Organization (NRAO) eNews**
Digital newsletter feature: “Molecular gas in high redshift galaxies”, **Jeff Shen** and Allison Man. https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas

Outreach

- 2019 - Present **Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast
Curate, write and record stories about the latest research and news in astronomy.
Episodes 177 - present.

Technical Skills

Data: Stan, R, Python, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, SQL
Other: Rust, BLAS/LAPACK, Astropy, L^AT_EX, regex, bash, vim, git, Docker, AWS, SymPy

Volunteering

- 2015 - 2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest.